



04. Mongoose Schema & Model Creation



Step 1: Defining Mongoose Schema

Schema define karta hai ki documents ka shape, data types aur defaults kya honge:

```
models/user.js - Schema Definition

const mongoose = require("mongoose");

const userSchema = new mongoose.Schema({
  name: String,
  email: String,
  age: Number,
  isActive: { type: Boolean, default: true }
});
```



Step 2: Creating Model Constructor

Model ek constructor class hoti hai jisse hum documents instantiate aur query karte hain:

```
models/user.js - Model Creation

// mongoose.model(modelName, schema)
const User = mongoose.model("User", userSchema);
module.exports = User;
```



Creating & Saving Documents (.save())

```
index.js - Save Document

// 1. In-memory JS instance bana (DB me save nahi hua)
const user1 = new User({
  name: "Amit Sharma",
  email: "amit@gmail.com",
  age: 21
});

// 2. .save() async function hai jo MongoDB me insert karta hai
user1.save()
  .then((res) => console.log("Saved to DB:", res))
  .catch((err) => console.log(err));
```



Direct Bulk Insert: Model.insertMany()

User.insertMany([...]) bina .save() call kiye seedha database me documents persist kar deta hai!



Pluralization Magic & Operation Buffering



Behind The Scenes Magic: Collection Pluralization

Jab aap model name **"User"** dete hain, Mongoose automatically usko **lowercase** aur **plural** bana kar MongoDB collection create karta hai:

"User" → **"users"** | **"Product"** → **"products"** | **"Person"** → **"people"**

Mongoose internally *pluralize* library use karta hai jo irregular English words (jaise Person → people) ko bhi accurately handle karti hai!



Operation Buffering (Interview Favorite Concept!)

Operation Buffering kya hai? Agar aapka database connection establish hone me 2 seconds lag rahe hain, tab bhi aap turant `User.find()` ya `user.save()` call kar sakte hain!

1. Query Call
`User.find()`

→ buffered in →

2. Memory Queue
(Buffers commands)

→ on connect →

3. DB Execution
(Fires automatically)



index.js — Operation Buffering Example

```
// DB connect hone se PEHLE hi query trigger kar di:  
User.find().then((data) => console.log("Data fetched:", data));  
  
// Connection baad me execute ho raha hai:  
mongoose.connect("mongodb://127.0.0.1:27017/collegeDB");  
// Mongoose query ko crash nahi karega, wait karega aur connect hote hi run karega!
```



TOPPER TIP: Agar aap custom collection name chahte hain toh 3rd argument pass kar sakte hain: `mongoose.model("User", userSchema, "custom_users")`!